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EXTENDED-ABSTRACT

The Landscape of Auto-bidding Auctions: Value versus Utility Maximization

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CCS Concepts: • **Theory of computation** → **Algorithmic mechanism design**; **Computational advertising theory**.

Additional Key Words and Phrases: internet advertising; auto-bidding; value maximization; return on spend constraints; private information; mechanism design.

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EXTENDED ABSTRACT

Programmatic advertising—using automated algorithms to buy and sell online advertisements instead of manual orders—has become the prevalent trading method in the online advertising market. It is estimated that 88% of all online advertisements were traded programatically in 2019 and this figure is expected to keep increasing in the years to come. Automated bidders simplify advertisers' bidding process by asking advertisers for their goals and then procuring advertising slots on their behalf by bidding in different auctions. While there are many types of automated bidders, one common strategy is to maximize the total number of clicks (or conversions) subject to a return on spend (RoS) constraint. These automated bidders ask advertisers to specify a target ratio and then procure ad slots while guaranteeing that the RoS, which is given by the ratio of total value to total spend, is greater than the desired target.

The emergence of automated bidders as the predominant mode of procurement in online markets brings into question whether the mechanisms currently used in practice are still effective in this new landscape. For example, most online advertisements are currently allocated using second- and first-price auctions, which do not explicitly take into account advertisers' objectives and RoS constraints. This opens the possibility for sellers to redesign their selling mechanisms to account for the goals of advertisers. Thus motivated, we study the problem of characterizing revenue-maximizing mechanisms for selling ads to advertisers with RoS constraints. Our goal is to understand the impact of the advertisers' private information and their objectives on the seller's

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revenue, and determine whether the first-best revenue, which is the optimal revenue when all the private information is public, is achievable.

We consider two objectives for the buyers (i.e., the advertisers): *value maximization* and *utility maximization*. Under value maximization, buyers seek to maximize the total value, which is given by the sum of predicted clicks or conversion, without taking into account for payments in the objective. Payments enter indirectly on the buyers' optimization problem through the RoS constraints, which imposes that the total payments should be at most a constant fraction of the total value. Utility maximization, in which buyers seek to maximize the difference between the total value and the total payments, is a standard choice for modeling buyer behavior in economics.

Moving to the information structure, we assume that either the buyers' target ratios or values are private. We first consider the case when *values are public and target ratios are private*. While it is typically assumed that advertisers' values are private, many advertisers increasingly rely on sophisticated prediction algorithms that are developed by the advertising platforms to predict clicks and conversions. These prediction algorithms have the benefit of being trained across many advertisers and long periods of data, are designed to generalize well, and give accurate predictions across different advertisers and market conditions. From this perspective, it is reasonable to assume that advertisers' values are public instead of private. On the other hand, the target ratio for the RoS constraint, which can be interpreted as the minimum return on spend an advertiser is willing to accept, depends on how advertisers internally evaluate an advertising campaign relative to other marketing channels or based on their internal accounting. This leads to buyers' target ratios being private from the perspective of the seller. We then move to the case when *values are private and target ratios are public*, which is the standard model in mechanism design. This model is appropriate when facing experienced advertisers or more sophisticated automated bidders who determine their own value predictions based on the user information shared by the seller.

An important feature of our model is that buyers aim to satisfy their RoS constraints in expectation. In practice, automated bidders participate in multiple repeated auctions and the RoS constraints need to be satisfied on average over the many auctions that the bidders participate. We abstract away from these dynamics and consider a single-period problem with in-expectation constraints, which is more tractable and sometimes leads to closed-form solutions. We model our problem as a sequential screening problem in which the seller first screens the buyers with respect to their target ratios (when these are private) and then, conditional on the reported targets, chooses an allocation and payment rule that screens buyers based on their valuations (when these are private). The buyers' reports are made to satisfy the RoS constraint in expectation over all possible values for all buyers. The objective of the seller is to maximize his revenue as given by the sum of the buyers' payments minus his cost.

Our goal in this paper is to study the design of revenue-maximizing mechanisms under different objectives and information structures described. When buyers are value maximizers, we prove that the first-best revenue is achievable when either the target ratios or the values are private. We further show that when both values and target ratios are private, first-best revenue is, in general, not achievable. In contrast, when buyers are utility maximizers, the first-best revenue is never achievable. Our main contribution is to provide a novel characterization of incentive compatibility for our sequential screening problem and then leverage this result to characterize optimal mechanisms in the settings described above.

We remark that many of our modeling assumptions, such as values being public or buyers maximizing total values, are non-standard in the literature. In proposing these models we hope to bring forward some new challenges that arise as advertisers increasingly adopt automate bidders and take a first step toward mapping the landscape of "auto-bidding auctions."